Reproducing and Extending Stochastic Switching in a Probabilistic Bandit Task Using Stickiness-Based Decision Models

Gillian Macdonald, Lenka Okasova, Bob Rice, and Ningqian Yang*
School of Mathematical Sciences, University of Nottingham, UK

Adaptive decision-making in uncertain environments requires integrating reward history to guide flexible action selection. Beron et al. (2022) [1] showed that mice performing a probabilistic two-choice bandit task exhibit stochastic, history-dependent switching behaviour that is accurately captured by reinforcement learning and recursive logistic regression models with choice history bias. Here, we reproduce the key behavioural and generative modelling results under the original 80-20 reward condition and extend the analysis to a noisier 70-30 environment. We compare the behaviour of the recursive logistic regression (RFLR) model and hidden Markov model (HMM) using reward-based performance metrics, including reward rate and reward-per-switch efficiency. We introduce nonlinear stickiness mechanisms by implementing temporally gated and surprise-dependent choice history bias and assess whether dynamic stickiness improves model fit and switching efficiency. Model performance is assessed using likelihood-based and behavioural metrics. Our results demonstrate how adaptive stickiness mechanisms influence switching efficiency and behavioural flexibility and provide further insight into the computational principles underlying decision-making in uncertain environments.

INTRODUCTION

Behavioural flexibility, the capacity to adjust behaviour in response to changing environmental contingencies, is a fundamental component of adaptive decision making. This capacity is essential in natural environments, where reward availability is often uncertain and subject to change. Impairments in behavioural flexibility are associated with several neuropsychiatric disorders, highlighting the importance of understanding the computational mechanisms that support adaptive decision making (Izquierdo et al., 2017) [2]. Probabilistic reversal learning (PRL) tasks provide a controlled framework for studying behavioural flexibility. In these tasks, reward contingencies reverse unpredictably without explicit cues, requiring agents to infer environmental changes from noisy reward feedback. Because individual outcomes are probabilistic, optimal behaviour depends on integrating information across multiple trials rather than relying on deterministic win-stay/lose-switch strategies. Computational accounts of reversal learning have broadly focused on two classes of models: reinforcement learning and inference-based approaches (Daw et al., 2005[3]; Wilson et al., 2014[4]). Reinforcement learning models update action values incrementally based on reward prediction errors, whereas inference-based models estimate latent environmental states governing reward contingencies. These approaches make distinct predictions about behavioural adaptation and switching dynamics. Beron et al. (2022)[1] investigated these mechanisms in mice performing a dynamic two-choice bandit task. They found that switching behaviour was graded, stochastic, and strongly dependent on recent reward history. Importantly, behaviour was accurately captured by reinforcement learning and recursive logistic regression models incorporating stochastic action selection and choice history bias. These findings suggest that adaptive decision making may be governed by history-dependent evidence integration rather than deterministic switching rules. This project reproduces the principal behavioural and computational findings of Beron et al. (2022)[1], with particular emphasis on switching dynamics and adaptation to contingency reversals. We further extend this framework by examining model behaviour in more uncertain reward environments and introducing nonlinear stickiness mechanisms that dynamically modulate choice history bias. These extensions aim to provide deeper insight into the computational principles underlying stochastic action switching and behavioural flexibility.

We hypothesised that models incorporating adaptive stickiness mechanisms would better capture switching behaviour and maintain behavioural efficiency under increased environmental uncertainty. We further predicted that increasing environmental stochasticity would disproportionately impair inference-based models relative to stickiness-based models.

METHODS: BERON ET AL. (2022)

Behavioural Findings Reported in Beron et al. (2022)

Mice exhibited graded, stochastic switching rather than deterministic responses to unrewarded trials. The conditional switching probability, $P(\text{switch} \mid \text{history})$, increased with accumulating negative evidence but remained bounded below one, indicating probabilistic rather than deterministic switching. Despite the stochasticity, mice achieved near-maximal reward rates, demonstrating efficient adaptation under uncertainty.

Computational Models

The authors compared models differing in how action values and environmental uncertainty were represented:

Win–Stay/Lose–Switch (WSLS). A heuristic strategy that repeats rewarded actions and switches after unrewarded outcomes. This model predicts deterministic switching and fails to capture graded behavioural responses.

Reinforcement Learning (RL). Model-free updating of action values using reward prediction errors with stochastic action selection. RL captures gradual adaptation but does not explicitly represent latent environmental states.

Inference-Based Models (HMM). A latent-state model that infers the current reward contingency from past outcomes. Switching is driven by belief updates, producing sharper transitions but overestimating switching behaviour relative to mice.

Recursive Logistic Regression (RFLR). A reduced model that integrates past choice-reward history with a bias toward repeating previous actions ("stickiness"). This model accurately reproduced switching behaviour and reward efficiency using a small number of parameters. Overall, this study concluded that mouse behaviour reflects stochastic, history-dependent evidence integration best captured by models incorporating stochastic policies and choice history bias.

METHODS: EXTENSIONS

Behaviour under increased uncertainty (70–30 condition):

Cross-Validated parameter estimation across reward environments: Following the modelling framework of Beron et al. (2022)[1],we fit the recursive formulation of logistic regression (RFLR) separately in the 80-20 and 70-30 reward environments. To ensure robust parameter recovery and prevent session-level leakage, a 5-fold cross-validation was performed at the session level. Sessions were randomly split into five folds with fixed random seed. For each fold, parameters α, β, and τ were estimated on four folds via maximum likelihood estimation. Optimisation was performed using stochastic gradient descent (10 000 steps, learning rate = 0.01), and log-likelihood was evaluated on the held-out fold. Final parameter estimates for each environment were computed as the mean across folds. This procedure therefore extends the fitting approach by explicitly validating generalisation across held-out sessions.

Generative model simulations: Using cross-validated parameters, we generated synthetic behaviour in both reward environments under three model classes: RFLR, Ideal HMM and Sticky HMM. All simulations were performed on the true latent state sequences extracted from experimental data to maintain identical block-transition structure across models. Reward delivery probabilities were set to 0.8/0.2 or 0.7/0.3, respectively, with transition probabilities estimated from empirical block statistics.

Behavioural metrics: To evaluate model performance at the macro-behavioural level, we quantified the trade-off between reward acquisition and behavioural cost using an efficiency frontier analysis. For each session, we computed total reward, total number of switches, and the reward-per-switch ratio. This extends the reward-per-switch analysis presented in the original paper [1] by enabling direct comparison of behavioural efficiency across environments with differing levels of reward uncertainty. To characterise adaptation dynamics at the micro-behavioural level, we analysed event-triggered switching probability aligned to block reversals. Switching probability was computed within a window between five trials before and ten trials after each contingency reversal. This approach mirrors the block-aligned $P(\text{switch})$ analysis shown in the original work [1], but extends it by explicitly contrasting switching dynamics between the 80-20 and 70-30 reward conditions, thereby quantifying environment-dependent differences in cognitive lag. Finally to asses cross-environment behavioural adaptation, we compared cross-validated parameter estimates (α, β, and τ) between the two reward conditions. In particular, we tested wether adaptation was primarily mediated through modulation of the stickiness parameter α, consistent with the environmental adjustments reported by Beron et al. (2022)[1].

Non-linear stickiness:

To extend the computational framework beyond fixed stickiness, we implemented nonlinear stickiness mechanisms incorporating both temporal and informational gating. These extensions were applied to the RFLR, HMM and forgetting Q-learning (F-Q) models. Temporal gating was implemented using an N-step commitment mechanism. Following each behavioural switch, the model's policy was constrained to repeat the new choice for a fixed number of trials (N), preventing immediate reversal. This manipulation models a persistence window during which behavioural switching is temporarily suppressed. Informational gating was implemented using surprise-dependent modulation of stickiness. The effective stickiness parameter was dynamically adjusted according to recent prediction error:

$$\alpha_{\text{dynamic}} = \alpha_{\text{base}} e^{-\gamma E}$$

where E represents accumulated prediction error over the previous N trials and γ controls sensitivity to prediction error. This formulation reduces stickiness when recent

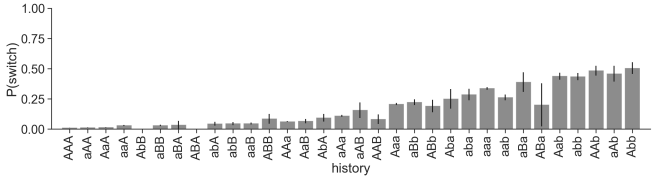


FIG. 1: **Nonparametric switching policy (mouse baseline).** Empirical conditional switching probabilities $P(\text{switch} \mid \text{history})$ for history length 3 computed directly from mouse behaviour (no model fit). Switch probability varies systematically with recent action–outcome history and remains bounded well below 1, consistent with graded and stochastic switching as reported by Beron et al and reproduced from our implementation of the dataset.

outcomes are surprising and increases persistence when predictions are reliable. Synthetic behavioural sessions were generated using the same probabilistic bandit task structure described previously. Each model was simulated across multiple sessions, and the dynamic stickiness parameter, switching frequency, and value estimates were recorded. These extensions allowed us to examine how adaptive stickiness mechanisms influence switching behaviour and learning dynamics under varying levels of environmental uncertainty. We hypothesised that dynamic stickiness would improve model fit to mouse behaviour and enhance switching efficiency by preventing excessive switching in response to stochastic reward fluctuations.

All simulations were implemented in Python using custom-written code. Synthetic behavioural data were generated for 5 independent sessions of 120 trials each. Reward contingencies evolved according to a two-state Markov process with transition probability P(switch) = 0.1. Model behaviour was simulated using the same task structure, and behavioural metrics were averaged across sessions. For nonlinear stickiness simulations, dynamic stickiness was modulated according to accumulated prediction error over N-step windows, with parameters matched to those used in the computational implementation.

Non-linear stickiness model

RESULTS

Reproduction of Beron et al. (2022)

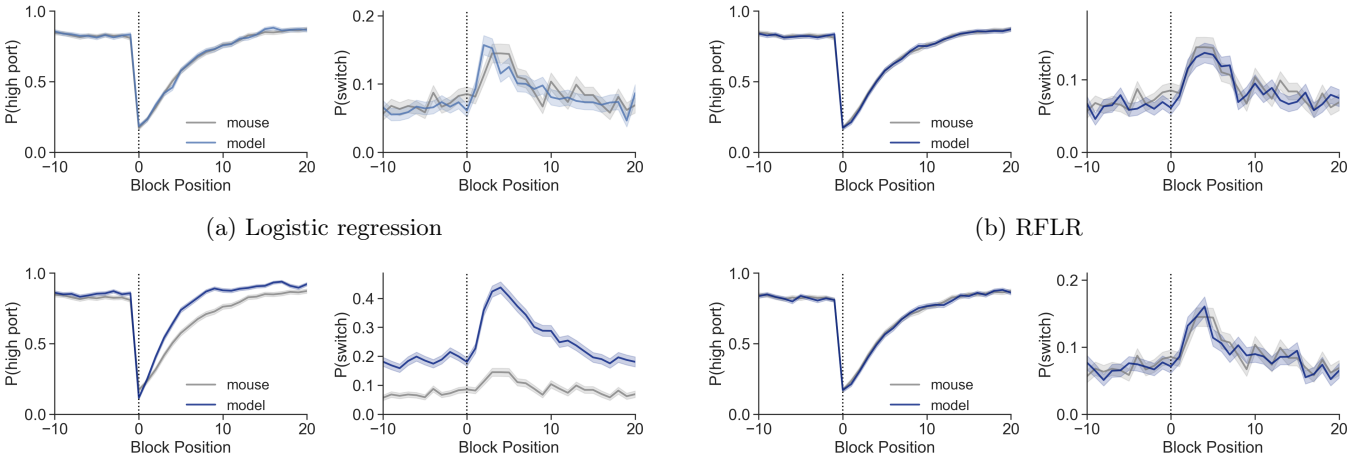
We first evaluated whether the implemented models reproduced the key behavioural signatures reported by Beron et al. Conditional switching probabilities computed directly from mouse behaviour exhibited graded dependence on recent reward history and remained bounded below one (Fig. 1), confirming that switching behaviour is stochastic rather than deterministic. Consistent with Beron et al. (2022), our reproduced model comparisons showed clear differences in their ability to capture behavioural dynamics. Logistic regression, RFLR, and forgetting Q-learning reproduced the gradual adaptation in both choice probability and switching behaviour around contingency reversals (Fig. 2). In contrast, the HMM exhibited sharper transitions and elevated switching rates relative to mice. These differences were also evident in the structure of conditional switching policies (Fig. 3). The RFLR closely matched the graded switching probabilities observed in mouse behaviour, whereas the HMM overestimated switching probability across many histories. Quantitative comparison confirmed this pattern: RFLR achieved the highest agreement with mouse switching probabilities (correlation = 0.93, SSE = 0.009), followed by logistic regression and forgetting Q-learning, while the HMM performed substantially worse (correlation = 0.72, SSE = 0.054; Table II). Scatter plots comparing model-predicted and mouse switching probabilities further illustrated the close correspondence for RFLR and larger deviations for the HMM (Figs. 4–5). Together, these results successfully reproduce the main findings of Beron et al., demonstrating that mouse behaviour is best explained by models incorporating stochastic evidence integration and choice history bias.

Validation of reproduction

These results confirm that our implementation accurately reproduces the key behavioural and modelling findings of Beron et al. (2022), validating the simulation framework for further extensions.

Extension: Behaviour under increased uncertainty (70–30 condition)

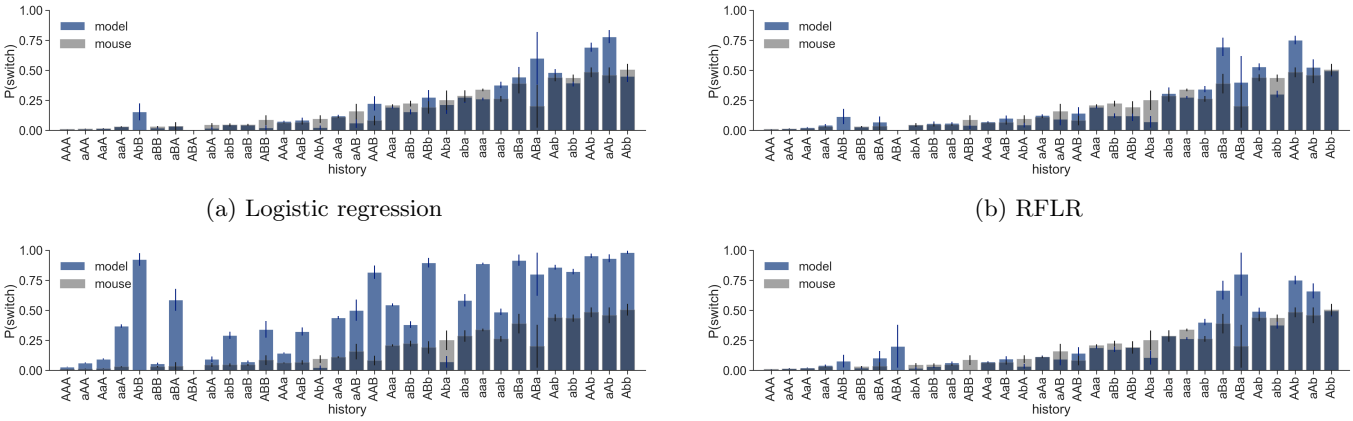
Modulation of parameters across environments: Cross-validated parameter estimates revealed systematic shifts between the 80–20 and 70–30 environments (Fig.6). In the 70–30 condition, the stickiness parameter (α) increased relative to the 80–20 condition, indicating greater behavioural persistence under increased reward stochasticity. By contrast, the inverse temperature parameter (β) and integration timescale (τ) exhibited comparatively smaller changes across environments. As illustrated in Fig.6, this shift in α constitutes the dominant cross-environment modulation, in line with the context-dependent parameter adjustments reported by Beron et al. (2022)[1]. These results suggest that behavioural adaptation to environmental uncertainty is mediated primarily through modulation of stickiness rather



(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

FIG. 2: **Reproduction: behaviour around contingency reversals.** Model (blue) vs mouse (grey) for P(high port) (left) and P(switch) (right) as a function of block position (dotted line at 0 denotes reversal). This reproduces the block-transition evaluation used in Beron et al. and enables qualitative comparison of how each model captures adaptation following reversals.



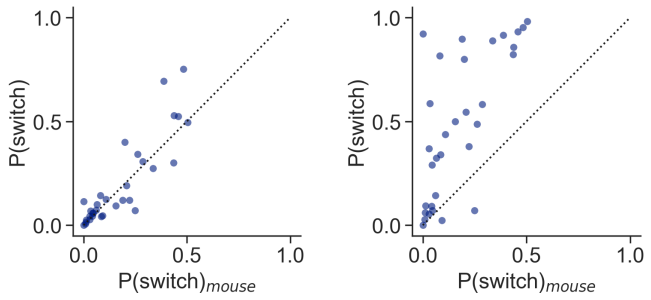
(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

FIG. 3: **Reproduction: conditional switching by recent history.** Mouse (grey) and model (blue) conditional switch probabilities P(switch | history) for history length 3 (histories sorted by mouse P(switch)). This reproduces the core "policy-shape" comparison in Beron et al. and highlights model-specific deviations from the empirical switching structure.

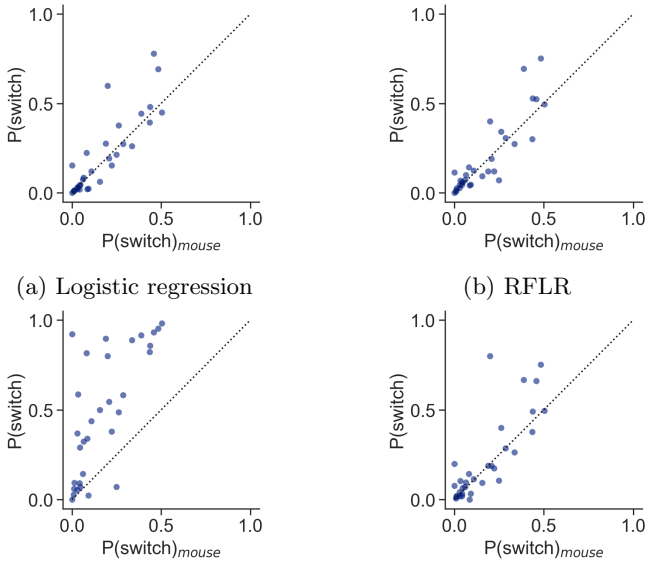
than alterations in reward sensitivity or evidence integration dynamics. Mice therefore appear to dynamically regulate perseverance as a function of environmental noise rather than relying on fixed decision parameters.

Ideal HMM over-switches under increased uncertainty: Model performance under the 70–30 condition is summarised in Table I and visualised in Figure 7. Although the Ideal HMM achieved a marginally higher reward rate than the RFLR (Table I), it has done so at the cost of substantially elevated switching frequency. The HMM exhibited approxiamtely three times more switches per session than both the mouse and RFLR model, resulting in a marked reduction in reward-per-switch efficiency. This trade-off is evident in Fig.7, where the HMM occupies a high-switch, lower efficiency regime relative to both the mouse behaviour and RFLR model. Under a stochastic policy, the inference-based HMM over-interprets unrewarded trials in the noisy 70–30 environment, leading to excessive switching despite near-maximal overall reward acquisition. Conversely, the RFLR model maintains switching behaviour much closer to the mouse regime (Table I), achieving comparable reward with substantially fewer behavioural switches. This confirms that stickiness-based mechanisms support robustness under increased reward uncertainty.



(a) RFLR (b) HMM

FIG. 4: **Reproduction: quantitative fit to conditional switching.** Scatter plots compare model-predicted vs mouse-observed $P(\text{switch} \mid \text{history})$ across histories (dotted line indicates identity). These plots provide a compact quantitative summary of agreement between empirical and model policies.



(a) Logistic regression (b) RFLR

(c) HMM (d) Forgetting Q-learning

FIG. 5: **Reproduction: quantitative agreement for all models.** Model-predicted vs mouse-observed $P(\text{switch} \mid \text{history})$ across histories (dotted identity line).

Cognitive lag preserves behavioural efficiency: Differences in switching dynamics are further clarified by the event-triggered analysis aligned to contingency reversals (Fig.8). In the 70–30 condition, the HMM exhibits rapid increases in switching probability immediately following reversals, reflecting heightened sensitivity to stochastic reward feedback. On the contrary, the RFLR and Sticky HMM display more gradual, graded switching, response consistent with structured evidence accumulation. Figure 9 directly compares empricial mouse switching dynamics across environments. Under the noisier condition (70–30), switching probability rises more slowly following a reversal than in the 80–20 condition, indicating increased cognitive lag under uncertainty. The RFLR reproduces this pattern, whereas the HMM demonstrates exaggerated switching peaks. This graded adaptation under noise prevents overreaction to isolated stochastic losses and preserves reward-per-switch efficiency (Table 1 and Fig.7). Therefore, the behavioural inertia that appears suboptimal in deterministic settings becomes functionally advantageous in noisy environments.

Statis parameters are environment specific: Additionally, cross-environment comparison of fitted parameters (Fig.6) demonstrates that optimal stickiness differs between reward conditions. Parameter settings tuned for one environment do not give equivalent efficiency when applied to the other, highlighting the brittleness of static strategies.

Extension: Non-linear stickiness mechanisms

We next examined how adaptive stickiness mechanisms influence model behaviour. Introducing surprise-dependent stickiness resulted in substantial reductions in effective stickiness across all models. Although the baseline stickiness parameter was set to $\alpha = 1.2$, the mean dynamic stickiness decreased to 0.351 in the RFLR model, 0.120 in the HMM, and 0.105 in the F-Q model. This demonstrates that prediction-error-dependent modulation effectively suppressed behavioural persistence when recent outcomes were inconsistent with model predictions. Temporal gating further stabilised behaviour by preventing rapid oscillatory switching. The mean locked switching rate was low in both the RFLR and HMM models (approximately 7%), indicating that the N-step commitment mechanism successfully reduced excessive switching. In the F-Q model, windowed updates produced more conservative value estimates. Final Q-values remained partially separated (left = 0.394, right = 0.499), reflecting slower but more stable learning dynamics under temporally gated updating. Together, these results demonstrate that adaptive stickiness mechanisms substantially alter behavioural dynamics by regulating the balance between persistence and flexibility.

DISCUSSION

This study reproduced and extended the findings of Beron et al. (2022), providing further insight into the computational mechanisms underlying stochastic action switching. Consistent with the original study, mouse behaviour exhibited graded, history-dependent switching that was accurately captured by the recursive logistic re-

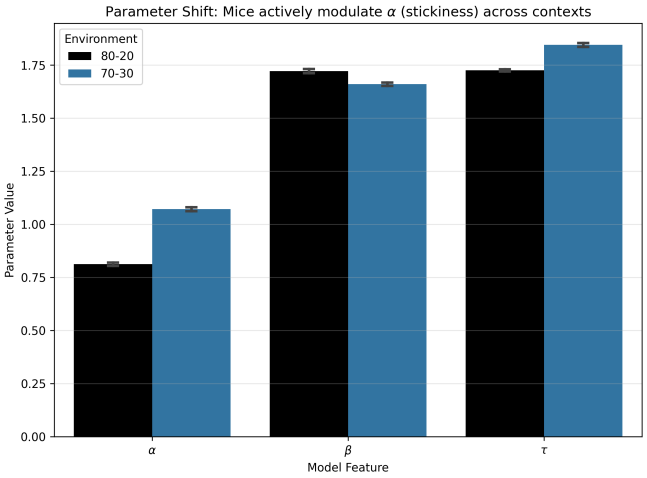


FIG. 6: **Context-dependent modulation of model parameters across reward environments.** Estimated parameter values for stickiness (α), inverse temperature (β), and integration timescale (τ) in the 80–20 and 70–30 reward conditions. Increasing environmental uncertainty was associated with a marked increase in effective stickiness (α) and integration timescale (τ), indicating greater behavioural persistence and slower evidence integration under noisy reward feedback. In contrast, inverse temperature (β) showed only modest variation across conditions. Error bars denote standard error across sessions. These results demonstrate that behavioural parameters adapt systematically to environmental uncertainty, supporting the hypothesis that stickiness is dynamically regulated to stabilise behaviour under stochastic reward conditions.

TABLE I: **70–30 extension: model performance comparison.** Summary statistics comparing mouse behaviour, RFLR, and HMM in the 70–30 reward condition. Reward rate quantifies overall performance, switches/session measures switching frequency, and reward-per-switch reflects switching efficiency. The HMM achieves slightly higher reward rate but switches excessively, resulting in substantially lower switching efficiency.

Model	Reward/session	Reward/switch	Switches/session
Mouse	0.595	11.960	51.6
RFLR	0.595	8.546	55.9
HMM	0.612	2.652	181.6

gression (RFLR) model but less well described by the hidden Markov model (HMM). In particular, the RFLR more closely matched the structure of conditional switching probabilities and behavioural adaptation around contingency reversals, supporting the interpretation that

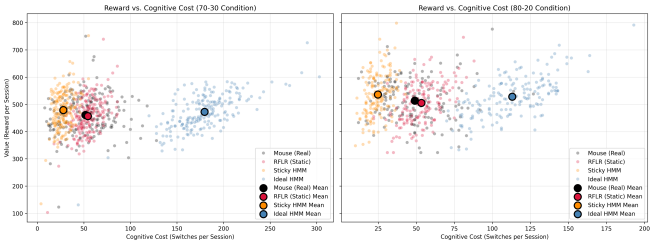


FIG. 7: **Trade-off between reward acquisition and switching cost across models and reward environments.** Relationship between total reward obtained per session and cognitive cost, defined as switching frequency, for mouse behaviour and model simulations in the 80–20 and 70–30 reward conditions. Each point represents an individual session, with larger markers indicating mean values. Under increased environmental uncertainty (70–30 condition), the hidden Markov model (HMM) exhibited substantially higher switching frequency without corresponding increases in reward, resulting in reduced reward efficiency. In contrast, the recursive logistic regression (RFLR) model maintained switching frequency and reward acquisition closer to mouse behaviour. These findings demonstrate that stickiness mechanisms improve behavioural efficiency by reducing excessive switching under stochastic reward conditions.

TABLE II: Model comparison summary (reproduction).

Model	Correlation (cprobs)	SSE
Logistic regression	0.91	0.012
RFLR	0.93	0.009
HMM	0.72	0.054
Forgetting Q-learning	0.90	0.015

mouse behaviour reflects stochastic evidence integration with choice history bias rather than optimal latent-state inference.

Mechanistically, these results demonstrate that stickiness need not be fixed but can be dynamically regulated according to environmental feedback. Surprise-dependent modulation allows the agent to reduce persistence when recent outcomes contradict expectations, facilitating behavioural flexibility. Conversely, temporal gating prevents rapid, noise-driven switching by enforcing short-term commitment following behavioural changes. These mechanisms address complementary limitations of the original models. Fixed stickiness improves stability but can impair flexibility when environmental contingencies change. In contrast, adaptive stickiness enables flexible adjustment of persistence according to environmental volatility. These findings suggest that behavioural stickiness may function as a dynamically regulated control parameter rather than a fixed behavioural

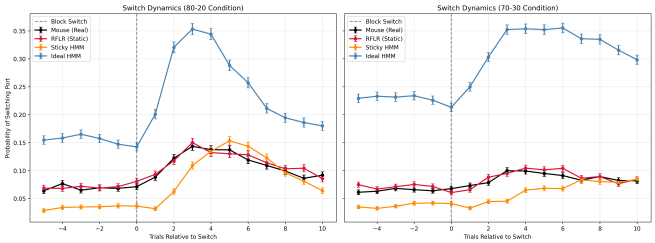


FIG. 8: **Trade-off between reward acquisition and switching cost across models and reward environments.** Relationship between total reward obtained per session and cognitive cost, defined as switching frequency, for mouse behaviour and model simulations in the 80–20 and 70–30 reward conditions. Each point represents an individual session, with larger markers indicating mean values. Under increased environmental uncertainty (70–30 condition), the hidden Markov model (HMM) exhibited substantially higher switching frequency without corresponding increases in reward, resulting in reduced reward efficiency. In contrast, the recursive logistic regression (RFLR) model maintained switching frequency and reward acquisition closer to mouse behaviour. These findings demonstrate that stickiness mechanisms improve behavioural efficiency by reducing excessive switching under stochastic reward conditions.

bias. This interpretation is consistent with the hypothesis that animals adjust their persistence in response to environmental uncertainty to optimise reward acquisition.

CONCLUSION

This study successfully reproduced and extended the computational findings of Beron et al. (2022), providing further insight into the mechanisms underlying stochastic switching behaviour in uncertain environments. Consistent with the original study, mouse behavioural dynamics were accurately captured by the recursive logistic regression model, whereas the hidden Markov model exhibited excessive switching due to its sensitivity to stochastic reward outcomes. Extending the task to a more uncertain 70–30 reward environment demonstrated that stickiness mechanisms confer behavioural robustness, enabling more efficient reward accumulation by preventing excessive switching. Further extension using nonlinear stickiness mechanisms showed that adaptive modulation of stickiness allows models to dynamically balance persistence and flexibility. Surprise-dependent stickiness reduced behavioural persistence when recent outcomes were unreliable, while temporal gating stabilised behaviour by preventing rapid, noise-driven switching. Together, these findings demonstrate that stickiness plays a fundamental computational role in adaptive decision-

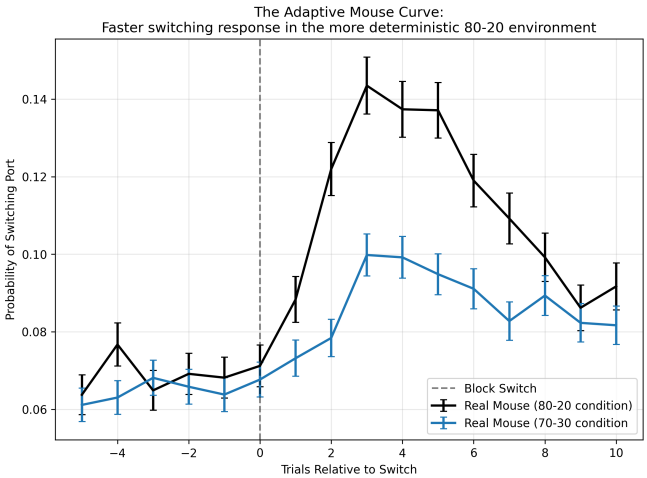


FIG. 9: **Mouse switching behaviour across reward environments.** Conditional switching probability as a function of trial position relative to contingency reversal in the 80–20 and 70–30 reward conditions. Switching probability increased more rapidly and reached higher peak values in the more deterministic 80–20 condition, whereas switching was slower and reduced in the more uncertain 70–30 condition. This pattern demonstrates adaptive modulation of switching behaviour according to environmental uncertainty, supporting the interpretation that behavioural stickiness is dynamically regulated to balance flexibility and stability.

making. Rather than acting as a fixed behavioural bias, stickiness may function as a dynamically regulated control parameter that enables efficient behaviour in stochastic environments. These results support the broader view that simple, history-dependent mechanisms can approximate optimal performance while maintaining robustness to environmental uncertainty.

---

* All figures were generated using custom Python code written by the authors, unless stated otherwise, e.g., for reproduction of original results. The full analysis and figure-generation pipeline is available at `https://github.com/BOBRICE1/Stochastic-Switching-Bandits`.

[1] C. C. Beron, J. Nehoran, A. J. Moss, M. R. Veit, and M. R. Cohen, Proc. Natl. Acad. Sci. U.S.A. **119**, e2113961119 (2022).
[2] A. Izquierdo, J. L. Brigman, A. K. Radke, P. H. Rudebeck, and A. Holmes, Nat. Rev. Neurosci. **18**, 655 (2017).
[3] N. D. Daw, Y. Niv, and P. Dayan, Nat. Neurosci. **8**, 1704 (2005).
[4] R. C. Wilson and Y. Niv, PLoS Comput. Biol. **10**, e1003670 (2014).